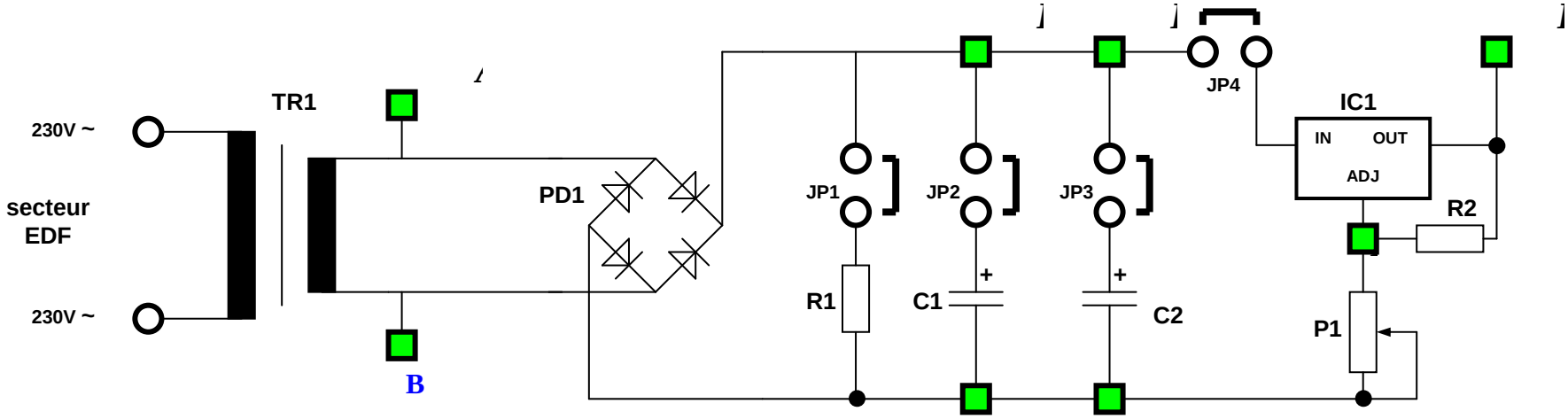
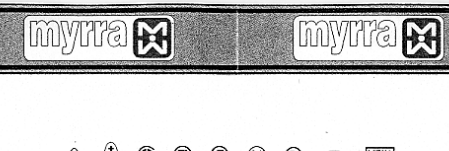
DESTRUCTEUR D'AIGUILLES SPAD

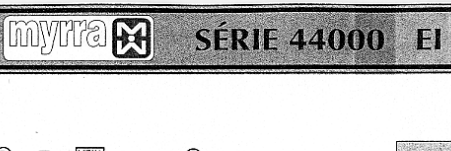


ANNEXE 1 : SCHEMA DE LA PLATINE D'ESSAIS

ANNEXE 2 : Le transformateur

3,2 VA EI 38-13,6 SÉRIE 44000





SÉRIE 44000 EI 38-13,6 3,2 VA

TENSION PRIMAIRE 117 V

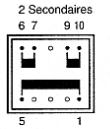
Protection Secondaire mA	Référence	Tension Secondaire V	Courant Secondaire m A	Tension à vide V	Température	Puissance
630	44205	6	533	8	T 70 B	3,2
400	44206	9	356	12	T 70 B	3,2
315	44207	12	267	16	T 70 B	3,2
250	44208	15	213	20	T 70 B	3,2
200	44209	18	178	24,1	T 70 B	3,2
160	44210	24	133	32,1	T 70 B	3,2
315	44211	2 x 6	2 x 267	2 x 8	T 70 B	3,2
200	44212	2 x 9	2 x 178	2 x 12	T 70 B	3,2
160	44213	2 x 12	2 x 133	2 x 16	T 70 B	3,2
125	44214	2 x 15	2 x 107	2 x 20	T 70 B	3,2
100	44215	2 x 18	2 x 89	2 x 24	T 70 B	3,2
80	44216	2 x 24	2 x 67	2 x 32,1	T 70 B	3,2

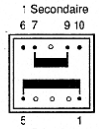
EN 60742

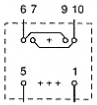
- Remplissage sous vide
- Carcasse à deux compartiments
- Plastiques autoextinguibles UL 94 VO
- Indice de protection IP 00
- Poids 150 grammes
- Résine classe B CEI 85 (essai 20 000 h CEI 216)
- Protection par fusible au secondaire (voir tableau) à assurer par l'utilisateur

- Version 30 V et 36 V homologuées VDE EN 60742 (fabrication sur demande)
- Tension d'isolement 4 KV (test MYRRA 6 KV)
- Matériel testé à 100 %
- Homologation : Procédure CCA sur demande

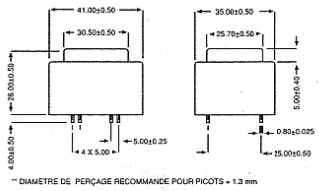
2 x 24 V non homologué transformateur de sécurité, conforme EN 60742 en tant que transformateur de séparation de circuits.







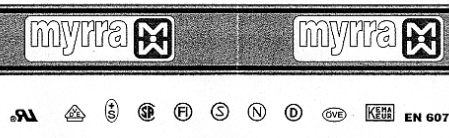
Implantation recommandée pour transformateurs à 1 secondaire (Permet l'utilisation d'un transformateur à 2 secondaires)

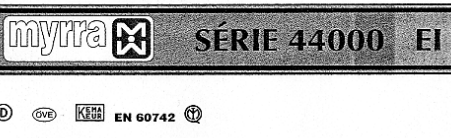


TENSION PRIMAIRE 230 V

Protection Secondaire mA	Référence	Tension Secondaire V	Courant Secondaire m A	Tension à vide V	Température	Puissance
630	44193	6	533	8	T 70 B	3,2
400	44194	9	356	12	T 70 B	3,2
315	44195	12	267	16	T 70 B	3,2
250	44196	15	213	20	T 70 B	3,2
200	44197	18	178	24,1	T 70 B	3,2
160	44198	24	133	32,1	T 70 B	3,2
315	44199	2 x 6	2 x 267	2 x 8	T 70 B	3,2
200	44200	2 x 9	2 x 178	2 x 12	T 70 B	3,2
160	44201	2 x 12	2 x 133	2 x 16	T 70 B	3,2
125	44202	2 x 15	2 x 107	2 x 20	T 70 B	3,2
100	44203	2 x 18	2 x 89	2 x 24	T 70 B	3,2
80	44204*	2 x 24	2 x 67	2 x 32,1	T 70 B	3,2

5 VA EI 42-14,8 SÉRIE 44000





SÉRIE 44000 EI 42-14,8 5 VA

TENSION PRIMAIRE 117 V

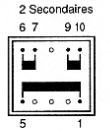
Protection Secondaire mA	Référence	Tension Secondaire V	Courant Secondaire m A	Tension à vide V	Température	Puissance
800	44241	6	833	8,4	T 50 B	5
630	44242	9	556	12,6	T 50 B	5
400	44243	12	417	16,9	T 50 B	5
315	44244	15	333	21	T 50 B	5
315	44245	18	278	25,3	T 50 B	5
200	44246	24	208	33,7	T 50 B	5
400	44247	2 x 6	2 x 417	2 x 8,4	T 50 B	5
315	44248	2 x 9	2 x 278	2 x 12,6	T 50 B	5
200	44249	2 x 12	2 x 208	2 x 16,9	T 50 B	5
160	44250	2 x 15	2 x 167	2 x 21	T 50 B	5
160	44251	2 x 18	2 x 139	2 x 25,3	T 50 B	5
100	44252	2 x 24	2 x 104	2 x 33,7	T 50 B	5

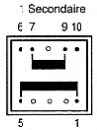
EN 60742

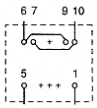
- Remplissage sous vide
- Carcasse à deux compartiments
- Plastiques autoextinguibles UL 94 VO
- Indice de protection IP 00
- Poids 200 grammes
- Résine classe B CEI 85 (essai 20 000 h CEI 216)
- Protection par fusible au secondaire (voir tableau) à assurer par l'utilisateur

- Version 30 V et 36 V homologuées VDE EN 60742 (fabrication sur demande)
- Tension d'isolement 4 KV (test MYRRA 6 KV)
- Matériel testé à 100 %
- Homologation : Procédure CCA sur demande

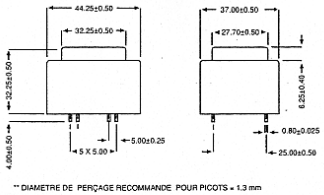
*2 x 24 V non homologué transformateur de sécurité, conforme EN 60742 en tant que transformateur de séparation de circuits.







Implantation recommandée pour transformateurs à 1 secondaire (Permet l'utilisation d'un transformateur à 2 secondaires)



TENSION PRIMAIRE 230 V

Protection Secondaire mA	Référence	Tension Secondaire V	Courant Secondaire m A	Tension à vide V	Température	Puissance
800	44229	6	833	8,4	T 50 B	5
630	44230	9	556	12,6	T 50 B	5
400	44231	12	417	16,9	T 50 B	5
315	44232	15	333	21	T 50 B	5
315	44233	18	278	25,3	T 50 B	5
200	44234	24	208	33,7	T 50 B	5
400	44235	2 x 6	2 x 417	2 x 8,4	T 50 B	5
315	44236	2 x 9	2 x 278	2 x 12,6	T 50 B	5
200	44237	2 x 12	2 x 208	2 x 16,9	T 50 B	5
160	44238	2 x 15	2 x 167	2 x 21	T 50 B	5
160	44239	2 x 18	2 x 139	2 x 25,3	T 50 B	5
100	44240*	2 x 24	2 x 104	2 x 33,7	T 50 B	5

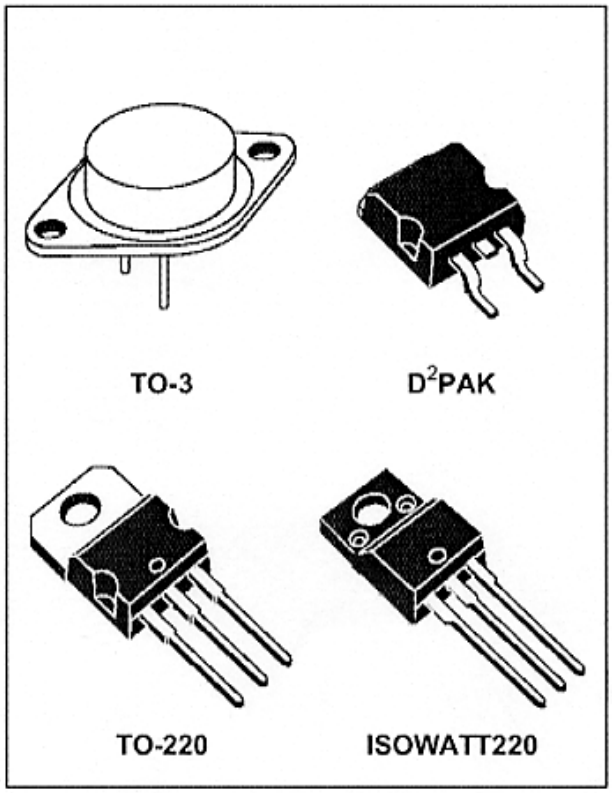
LA QUALITÉ EN SÉRIE

ANNEXE 3 : Le régulateur LM317

LM117/217
LM317

1.2V TO 37V VOLTAGE REGULATOR

- OUTPUT VOLTAGE RANGE : 1.2 TO 37V
- OUTPUT CURRENT IN EXCESS OF 1.5A
- 0.1% LINE AND LOAD REGULATION
- FLOATING OPERATION FOR HIGH VOLTAGES
- COMPLETE SERIES OF PROTECTIONS : CURRENT LIMITING, THERMAL SHUTDOWN AND SOA CONTROL



DESCRIPTION

The LM117/LM217/LM317 are monolithic integrated circuit in TO-220, ISOWATT220, TO-3 and D^2PAK packages intended for use as positive adjustable voltage regulators.

They are designed to supply more than 1.5A of load current with an output voltage adjustable over a 1.2 to 37V range.

The nominal output voltage is selected by means of only a resistive divider, making the device exceptionally easy to use and eliminating the stocking of many fixed regulators.

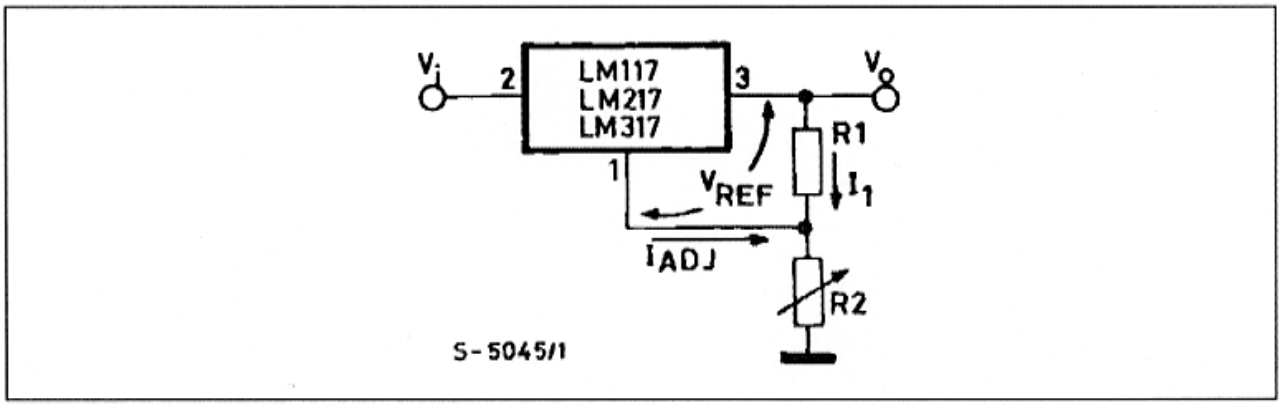
ABSOLUTE MAXIMUM RATING

Symbol	Parameter	Value	Unit
V_{i-o}	Input-output Differential Voltage	40	V
I_O	Output Current	Internally Limited	
T_{op}	Operating Junction Temperature for: LM117 LM217 LM317	-55 to 150 -25 to 150 0 to 125	°C °C °C
P_{tot}	Power Dissipation	Internally Limited	
T_{stg}	Storage Temperature	- 65 to 150	°C

THERMAL DATA

Symbol	Parameter		TO-3	TO-220	ISOWATT220	D^2PAK	Unit
$R_{thj-case}$	Thermal Resistance Junction-case	Max	4	3	4	3	°C/W
$R_{thj-amb}$	Thermal Resistance Junction-ambient	Max	35	50	60	62.5	°C/W

BASIC ADJUSTABLE REGULATOR



ELECTRICAL CHARACTERISTICS (V_i - V_o = 5 V, I_o = 500 mA, I_{MAX} = 1.5A and P_{MAX} = 20W, unless otherwise specified)

Symbol	Parameter	Test Conditions		LM117/LM217			LM317			Unit
				Min.	Typ.	Max.	Min.	Typ.	Max.	
ΔV_o	Line Regulation	V_i - V_o = 3 to 40 V	T_j = 25 °C		0.01	0.02		0.01	0.04	%/V
					0.02	0.05		0.02	0.07	%/V
ΔV_o	Load Regulation	$V_o \leq$ 5V I_o = 10 mA to I_{MAX}	T_j = 25 °C		5	15		5	25	mV
					20	50		20	70	mV
		$V_o \geq$ 5V I_o = 10 mA to I_{MAX}	T_j = 25 °C		0.1	0.3		0.1	0.5	%
					0.3	1		0.3	1.5	%
I_{ADJ}	Adjustment Pin Current				50	100		50	100	µA
ΔI_{ADJ}	Adjustment Pin Current	V_i - V_o = 2.5 to 40 V I_o = 10 mA to I_{MAX}			0.2	5		0.2	5	µA
V_{REF}	Reference Voltage (between pin 3 and pin 1)	V_i - V_o = 2.5 to 40 V I_o = 10 mA to I_{MAX} $P_D \leq P_{MAX}$		1.2	1.25	1.3	1.2	1.25	1.3	V
$\frac{\Delta V_o}{V_o}$	Output Voltage Temperature Stability				1			1		%
$I_{o(min)}$	Minimum Load Current	V_i - V_o = 40 V			3.5	5		3.5	10	mA
$I_{o(max)}$	Maximum Load Current	V_i - $V_o \leq$ 15 V $P_D < P_{MAX}$		1.5	2.2		1.5	2.2		A
		V_i - V_o = 40 V $P_D < P_{MAX}$ T_j = 25 °C			0.4			0.4		A
e_N	Output Noise Voltage (percentance of V_O)	B = 10Hz to 10KHz T_j = 25 °C			0.003			0.003		%
SVR	Supply Voltage Rejection (*)	T_j = 25 °C f = 120 Hz	C_{ADJ}=0		65			65		dB
			C_{ADJ}=10µF	66	80		66	80		dB

(*) CADJ is connected between pin 1 and ground.

Note:

(1) Unless otherwise specified the above specs, apply over the following conditions : LM 117 T_j = – 55 to 150°C; LM 217 T_j = – 25 to 150°C ; LM 317 T_j = 0 to 125°C.

Figure 1 : Output Current vs. Input-output Differential Voltage.

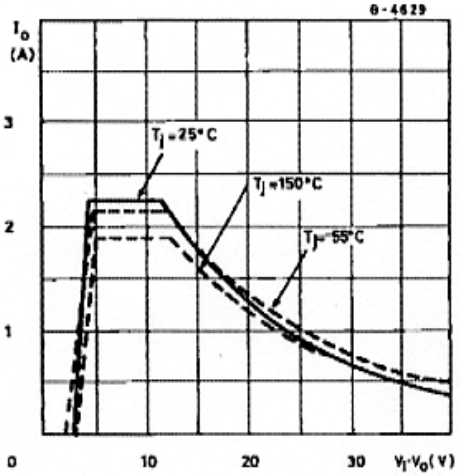


Figure 2 : Dropout Voltage vs. Junction Temperature.

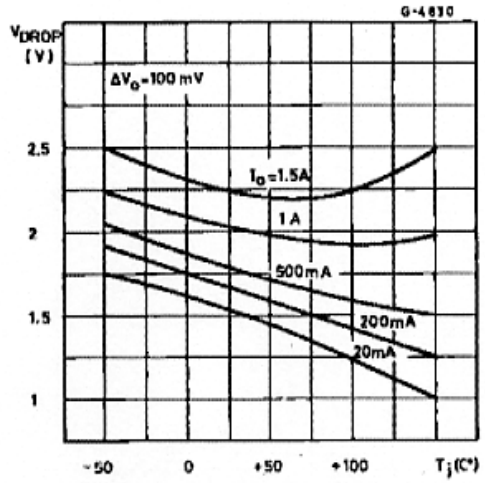


Figure 3 : Reference Voltage vs. Junction

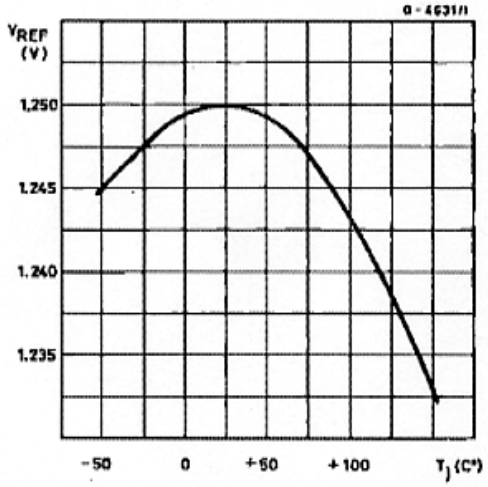
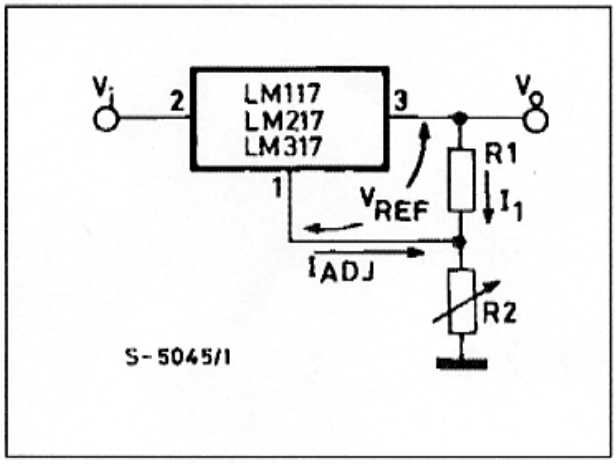


Figure 4 : Basic Adjustable Regulator.



APPLICATION INFORMATION

The LM117/217/317 provides an internal reference voltage of 1.25V between the output and adjustments terminals. This is used to set a constant current flow across an external resistor divider (see fig. 4), giving an output voltage V_O of:

$$V_O = V_{REF}\left(1 + \frac{R_2}{R_1}\right) + I_{ADJ} R_2$$

The device was designed to minimize the term I_{ADJ} (100µA max) and to maintain it very constant with line and load changes. Usually, the error term $I_{ADJ} \cdot R_2$ can be neglected. To obtain the previous requirement, all the regulator quiescent current is returned to the output terminal, imposing a minimum load current condition. If the load is insufficient, the output voltage will rise.

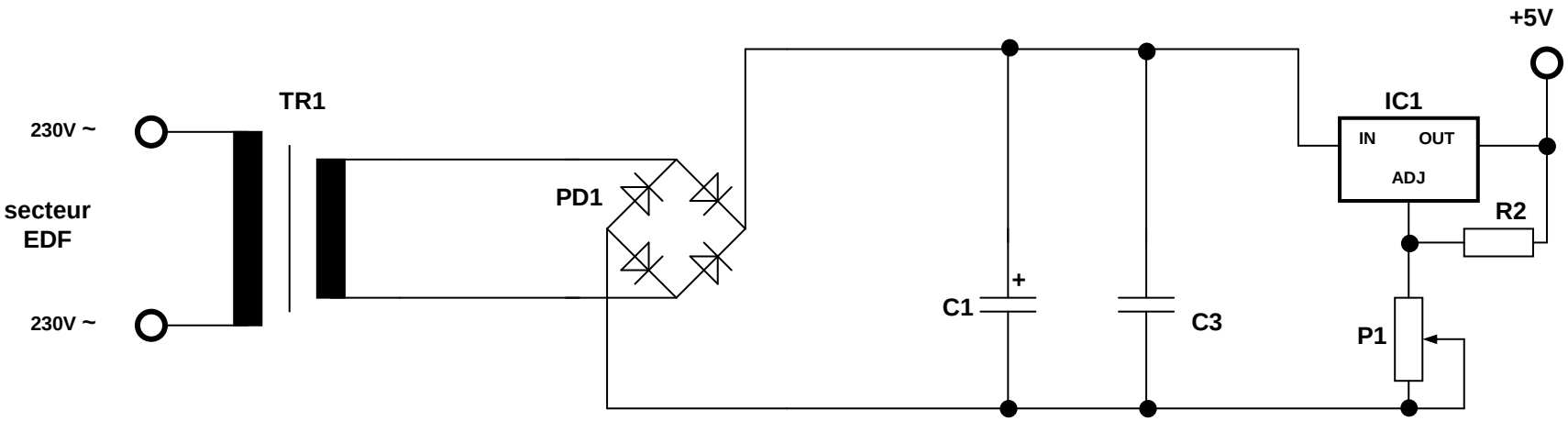
Since the LM117/217317 is a floating regulator and "sees" only the input-to-output differential voltage, supplies of very high voltage with respect to ground can be regulated as long as the maximum input-to-output differential is not exceeded. Furthermore, programmable regulator are easily obtainable and, by connecting a fixed resistor between the adjustment and output, the device can be used as a precision current regulator.

In order to optimise the load regulation, the current set resistor R1 (see fig. 4) should be tied as close as possible to the regulator, while the ground terminal of R2 should be near the ground of the load to provide remote ground sensing.

Performance may be improved with added capacitance as follow:

An input bypass capacitor of 0.1µF

An adjustment terminal to ground 10µF capacitor


+5V
TR1
IC1
230V ~
IN
OUT
ADJ
secteur
EDF
PD1
R2
+
C1
C3
P1
230V ~